Satyam Kulkarni

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Profiles

Satyam Kulkarni
LinkedIn

Education

MS - Computer Science
Woolf University
CGPA - 3.7/4
August 2023 - February 2025

BE - Mechanical Engineering
Pune University
CGPA - 8.31/10
July 2016 - May 2020

Skills

Technical Skills

Python, SQL, Optimization, Jupyter notebook, Pandas, Numpy, Scikit-learn, TensorFlow, PyTorch, Matplotlib, Seaborn, Feature Engineering, Neural Networks, Regression Analysis, Hypothesis Testing, Time Series Analysis, LLM's, Random Forest, PCA, NLP, Flask, Streamlit, Data Analysis, Power BI, Advanced Excel, Tabluea, Presentation, Statistics

Soft Skills

Analytical Thinking, Critical Thinking, Problem Solving, Communication Skills, Collaboration, Data Storytelling, Attention to detail, Business Acumen, Adaptability

Domain/Industry Knowledge

Automotive and Automobile, Finance, Banking, Manufacturing, E-commerce, Retail, Aerospace and Defense

Certifications

Full Stack Data Science and AI
IIT Guwahati
January 2024
Certificate

SQL - Advanced
HackerRank
January 2024
Certificate

Specialisation on Industry 4.0
Coursera
October 2020
Certificate

Achievements

HackerRank - Python: Silver Badge
Badge

HackerRank - SQL: Silver Badge
Badge

HackerRank - Problem Solving : Silver Badge
Badge

Summary

Computer Science graduate with 2+ years of experience in CAE simulation, analysis, and solving complex problems using data-driven approaches. Proficient in Python, SQL, Power BI, Advanced Excel, Tabluea and machine learning techniques, with expertise in extracting actionable insights by performing data analysis, building predictive models, and streamlining processes to enhance operational efficiency.

Work Experience and Transferable Skills

Catalog Automation Lead — **December 2024 - Present**
Amazon — Bangalore, Karnataka

- Managed data domains, analyzed performance metrics, and implemented configuration changes using Python, MySQL/MSSQL, and open-source technologies.
- Developed a scalable machine learning pipeline using Amazon SageMaker to predict customer satisfaction scores in an e-commerce setting, leveraging deep learning classification algorithms such as neural networks to analyze customer feedback data.
- Performed comprehensive data preprocessing, including text cleaning, feature engineering, and data augmentation, followed by model training and hyperparameter optimization using TensorFlow and PyTorch to improve prediction accuracy.
- Deployed the model on an Amazon SageMaker endpoint for batch predictions, enabling periodic customer satisfaction analysis and integration with the e-commerce platform for insights, while monitoring and retraining the model to adapt to evolving customer feedback.

CAE Engineer — **March 2021 - May 2023**
Cummins — Pune, Maharashtra

- **Collaboration and Communication:** Collaborated with cross-functional teams, including product, design, and engineering, to gather detailed requirements and provide actionable, data-driven insights. Effectively communicated complex simulation results and recommendations to non-technical stakeholders, translating technical findings into clear, understandable reports and powerpoint presentations. This ensured alignment with product development goals, informed decision-making, and supported continuous improvement in product designs and performance.
- **Preprocessing, Optimization, and FEA Model Validation:** Performed geometry cleaning and preprocessing in ANSYS, ensuring accurate analysis by identifying inconsistencies and removing unnecessary elements. Applied Design of Experiments (DoE) methodologies to optimize design parameters. Validated simulation results for accuracy and demonstrated the ability to select trade-offs between accuracy and computational efficiency in CAE processes. This hands-on experience in numerical methods and optimization techniques is directly applicable in Data Science and Analytics.
- **Numerical Modeling and Simulation**: Gained extensive knowledge and hands-on experience in numerical modeling and simulation, applying advanced mathematical principles and optimization techniques. This foundation is integral to solving complex problems in data science and Analytics.

Academic Projects

AI-Powered Financial Literacy Chatbot

- Developed an AI chatbot using Large Language Models (LLMs) to deliver personalized financial advice, addressing the financial literacy gap with insights from the Finance Alpaca dataset and curated financial articles.
- Fine-tuned the Llama-2-7B model with QLoRA, optimizing for efficiency and relevance, achieving strong conversational capabilities to answer complex financial queries and offer actionable guidance.
- Created a user-friendly HTML frontend integrated with a Flask backend, deployed to the Hugging Face repository, providing seamless, scalable access to personalized financial advice.

Retail Sales Prediction

- Developed an end-to-end machine learning model to forecast retail sales using the Rossman dataset, uncovering patterns to optimize store performance.
- Performed data preprocessing, including loading, cleaning, handling missing values, removing duplicates, and applying one-hot encoding and outlier removal. Used a RandomForest Regressor model to make predictions on retail sales, optimizing model performance and achieving an R-squared value of 0.88.
- Built a web application using Streamlit, and utilized Docker for containerization, Kubernetes for orchestration, and Jenkins with MLflow and Apache Airflow for CI/CD pipelines.